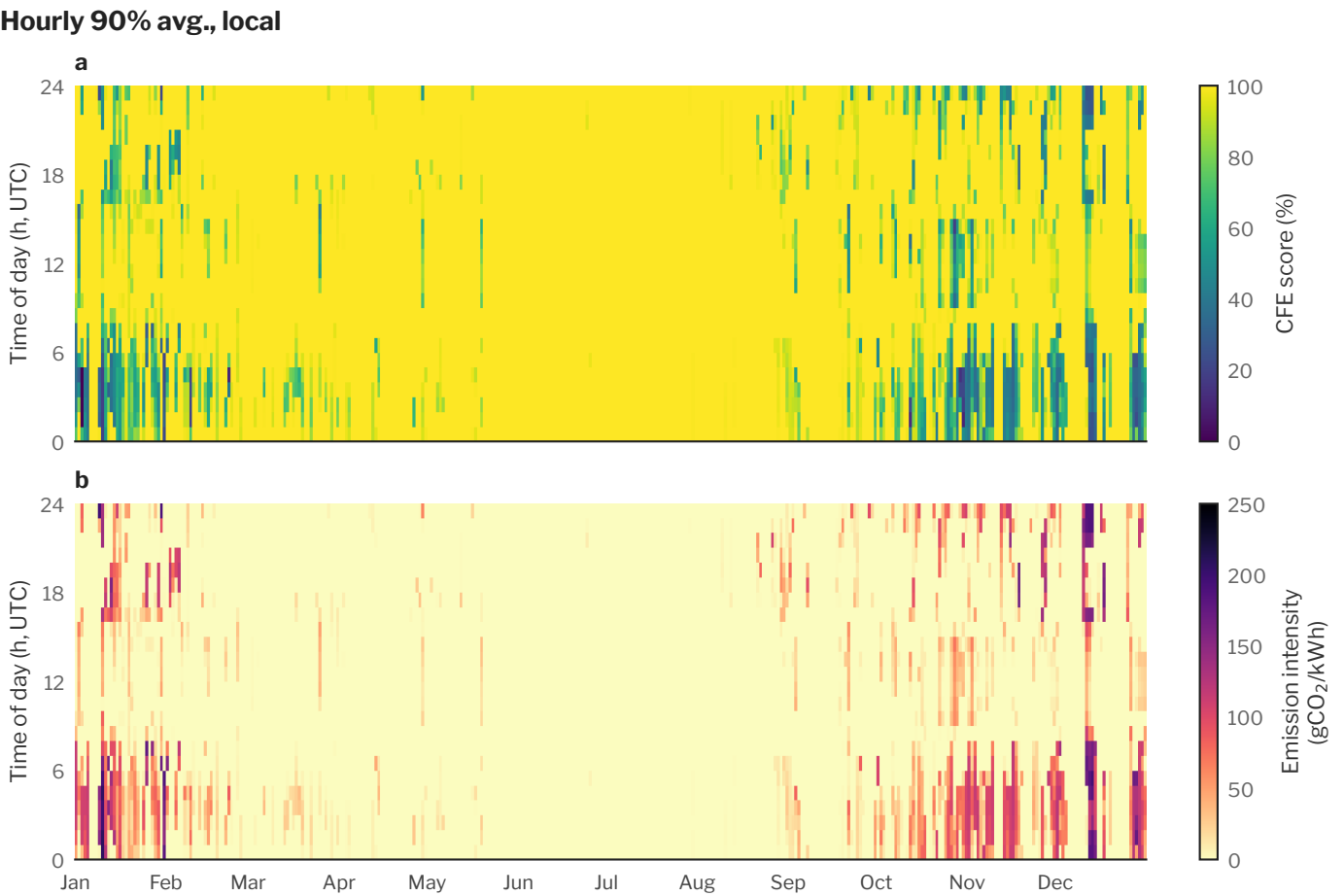

Hourly CFE score (a) and emission intensity (b) of the consumption of a 500 MW data center in Aragon, one cell per 1-hour snapshot (UTC). Local: PPAs in Aragon only. CFE score: the share of consumption covered by the data center's PPAs and battery plus the renewable share of its grid purchases (import-traced, nuclear excluded); annual mean 95%. Emission intensity: its grid purchases valued at the hourly emission factor of Spanish supply (imports traced to their exporter's mix), per kWh consumed; annual mean 8 gCO2/kWh.